The *Catalogue Astrochimique de Sections Efficaces d'Ionisation* (CASEI): A Database Of Low Energy Total Ionisation Cross-Section For Astrochemical Purpose

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Total Ionisation Cross Section (TICS) reprents a certainly important factor for a large number of physical domains.

1 Introduction

Electron-driven ionisation plays a big role in planetary and interplanetary medium. The result of it the production of reactive radicals. One of the main parameter to quantify this phenomena is the cross-section of ionisation of the molecule, either partial (Partial Ionisation Cross Section, **PICS**) or total (Total Ionisation Cross Section, **TICS**). While it can be mesured experimentally [34, 22, and reference therein], this is not necesseraly the case for every molecules. In fact, some molecules are either absent or under covered litterature-wise. Those include molecule of interest for planetary and exoplanetary science, like phosphine PH_3, found in the atmosphere of gas giants and in the interstellar medium [6, 28, 36], propaenitril C_3H_5N, found in Titan atmosphere (cite), or even brome bearing molecules such as ..., ..., ... The same can be said for isomers, as not all isomer of a define specie have the TICS calculated for all its isomer. It is, however, possible to compute this parameter. Total Ionisation Cross-Section (TICS) database established using computing method were already done, such as ALeCS [15], NIFS (citer) or NIST (citer). They, howerever, lack the presence of some of those *exotic* specie, or are either complicated to acces for general public.

In this work, I present a database of TICS for 274 neutral and stable molecules made with open-source tools. This database is also available on a website hosted on GitHub and is publicly available[1]. All the content of the website can be downloaded freely.

2 Methodology

The methodology employed in this work is based around the geometry of each molecules.

2.1 Molecular geometry pipeline

One of the most extensive database for chemical species is PubChem [23]. Practically all of the molecular species present in the database have 3D conformers available to download. The files associated to those conformers, `.xyz` files, consits in the space position of each atom in 3D space. The code employed in this work make a request at PubChem server with the name of the searched molecule and fetch for the 3D structure as for other properties such as the formula. The 3D conformer catalogue of Pubchem was, as their documentation suggests[2], calculated using advanced quantum computational methods of atomic-force fields repultation, with the assurance of a great accuracy between experimental, theoritical and modeled 3D conformers (see [5]). In this case, atomic coordinates should already be optimized and no further optimization is needed (see appendix **??**).

One problem emerge regarding molecules with no 3D conformer available. Those are presents in italic in table 1. While they represent a very small fraction of this database, and appears as *niche* astrochemically speaking at first glance, they do represents molecules with a certain degre of interest. For instance, Germane GeH_4 could be present in the atmosphere of giant planets [9, 29] as well as arsenic bearing molecules [4, 38]. First calculations were initally made using 2D structure, but it was quickly proven to be false. While it could be possible to get 3D conformers from other sources (as such, Jmol [17]), it is easier for automated workflow to keep the same source of data.

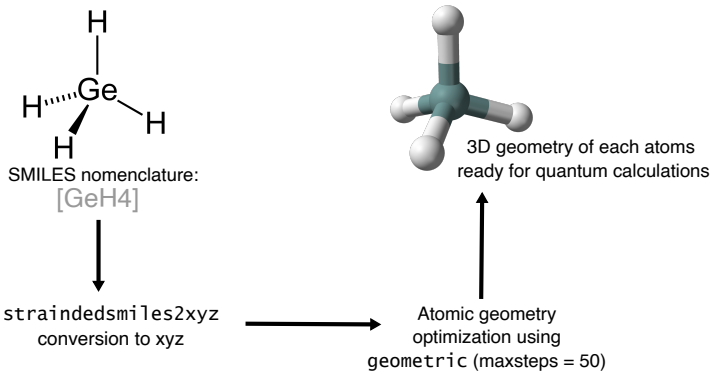


Figure 1: Overview of the molecular geometry pipeline for molecules without 3D conformer available. Germane GeH_4 is used as an example.

Thankfully, it is possible to obtain 3D geometry starting from PubChem 2D structures. For that, we used the SMILES id of each molecule. Using the `strainedSMILES2xyz` python package [10], we can retrieve a coherent 3D structure from the SMILES structure. While this packages is orignially aimed at strained ring system in complex molecules, it was proven to be very effective at converging to accurate 3D geometry (see the discussion section of [10]). It uses tools from the `RDKit` python library [27] at the start. Namely, the *Experimental-torsion-knowledge distance geometry* (ETKDG) method proposed by Riniker and Landrum [37] which relies on atomic distances in order to find equilibrum in force field from each atoms. While `strainedSMILES2xyz` uses ORCA quantum model [30] in order to optimize obtained geometry, we choosed to rely

[1] https://lh67000.github.io/casei

[2] https://pubchem.ncbi.nlm.nih.gov/docs/pubchem3d

on open-source tools to do so. As such, we used the `geometric` optimization package by Wang and Song [42] for several cycles, in order to get a geometry that takes into account atomic translations and rotations. This optimization is done during PySCF calculations using the Hartree-Fock (HF) method, as described in subsection 2.3. Figure 1 recapitulates the overall workflow employed for those molecules. The molecules that use this pipeline are indicated in grey in table 1. However, an exception was made for sulfur hexafluoride SF_6, as it was proven to be very difficult to converge to a stable geometry that was coherent. As such, and regarding its importance for astrochemistry, the 3D conformer of SF_6 was downloaded from Jmol and used for calculations in that specific case.

2.2 The BEB method for TICS calculation

The *Binary Encounter Bethe* (BEB) method was proposed by Kim and Rudd [24] in order to provide a simplier model than the *Binary Encounter Dipole* (BED) for ionisation cross-sections calculation. The BEB method estimates the cross-section of total ionisation using the electron energy of the Molecular Orbital (MO) and the kinetic energy of the electron whithin. In application, the BEB method can expressed as, for a distinct MO i:

$$\sigma_i(T) = \frac{S}{t+u+1}\left[\frac{\ln t}{2}\left(1-\frac{1}{t^2}\right)+1-\frac{1}{t}-\frac{\ln t}{t+1}\right] \quad (1)$$

with $t = \frac{T}{B}$, $u = \frac{U}{B}$, where T the ionisation particle energy and B and U are the aforementioned MO electrons and kinetic energy, and S is expressed as:

$$S = 4\pi a_0^2 N_i \left(\frac{R}{B}\right)^2 \quad (2)$$

with a_0 the Bohr radius (0.5292 $\mathring{A}$), R the Rydberg constant (13.6 eV) and N_i the number of electrons in the MO. For neutral stable species, N is always equal to 2. This, however, gives the cross section for only one MO. To get the total ionisation cross section, we need to sum all the $\sigma_i(T)$ of each MO, which gives us:

$$\sigma_{TICS}(T) = \sum_{i=1}^{n} \sigma_i(T) \quad (3)$$

Concerning T, this parameter is kept at relatively low energy. We considered 10^3 values in a logscale array with $T \in [10^1; 10^5]$ eV.

2.3 Quantum chemistry parameters calculations

As such, BEB calculations requires the MO electrons energies (B) as well as their kinetic energies (U) . While NIST database (cite) propose this parameter for several molecule, this is limited in terms of selection. One way is to calculate these parameter. To do that, PySCF quantum chemistry model [40] was employed. This open-source python library have been used in several published papers in the past (cite). Using the downloaded geometries, we can input them into the model and obtain the required paramaters. This model is based on the Self Consistent Field (SCF) method of quantum chemistry and resolves the HF solution. The basis-set family used in this work is the Karlsruhe `def2-TZVPPD` basis-set – valence triple-zeta with sets of polarization and diffuse functions [43, 35]. Although a quadruple-zeta (QVZ) basis set could have been used, we used a triple-zeta (TVZ) basis set in this work for computational reasons. This type of basis, in particular the `def2-TZVPP` basis, is proven to be reliable for HF modeling for a majority of the periodic table, from H up to Rn [18, 44]. Several BEB-related papers use either the cross-correlated Dunning Dunning [12] `cc-` base [15, 26] or Pope [25] gaussian `6-311G` basis [2, 21]. At our knowledge, this is the first use of a `def2-` basis for TICS calculations. Only Restricted Hartree-Fock (RHF) calculations were performed, as all molecules are neutral and stable and thus have closed shell.

B was obtained using Koopman's theorem, $B = \varepsilon$. U can be expressed as :

$$U = <\varphi|\hat{T}|\varphi> \quad (4)$$

The BEB method, by itself, has limitations for heavy elements, with quantum number $n > 2$ or $Z > 10$ [16, 19, 1, 39]. The discrepancy arises because the BEB method uses, as shown in Eq. 1, the kinetic energy of each molecular orbital as a variable. This is problematic for heavier elements, as inner electronic layers of this type have very high kinetic energies. However, ionisation should occur when the electron is far from the centre of the shell, i.e. when it is slower and thus has a lower kinetic energy. An overestimation of the kinetic energy leads to an underestimation of σ_{TICS}. The solution proposed by Huo and Kim [19], and employed in several BEB-related studies on heavy elements [16, 11, 39], is to use the effective core potential (ECP) method. An ECP limits the interaction of valence electrons with the core shell by replacing the latter with analytical functions that represents the interaction between the two. These functions appear as effective potential, while the valence electrons are described by pseudo-valence orbitals that do not penetrate the inner region of the nucleus. In a nut*shell*, the ECP enables the removal of high kinetic energies of core electrons from the BEB sum, which would otherwise artificially minimalize σ_{TICS}.

The ECP method can be used in PySCF and, in order to include as many molecules as possible in the database, was implemented in this work. This, however, requires a different basis than the all-electrons Karlsruhe `def2-`, because its related ECP basis only start at Rb ($Z \geq 37$) and thus don't work for lighter elements, *e.g* $3p$ and $4p$ elements. The only basis that has an ECP basis that supports these that is inherently available in PySCF and in the Basis Set Exchange database[3] [33] is the `crenbl` basis set [20, 14]. Thankfully, the open-source Pseudopotentiallibrary[4] provides modern Dunning derived basis combo for both $3p$ and $4p$ elements. For Al, Si, P, S, Cl and Ar bearing molecules, we used the `aug-cc-pVTZ` orbital basis along with its associated ECP basis proposed by [3], whereas for Ge, As, Se and Br bearing molecules we used the same orbital basis and the ECP basis proposed by [41]. The impact of ECP on TICS calculation is discussed in subsection 3.2.

3 Results

3.1 Overview

260 total ionisation cross section of unique molecules have been calculated, and up to 10 isomer for some formula[5]. The complete list is visible on table 1. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus

[3] https://basissetexchange.org/
[4] https://pseudopotentiallibrary.org/
[5] Like C_6H_{12} where 10 isomers are available.

Table 1: List of molecules present in the database. Superscript number indicates the number of isomers. Grey molecules indicates that no 3D conformer was available on PubChem and that the geometry was generated from SMILES structure.

Number of atoms													
1	2	3	4	5	6	7	8	9	10	11	12	13	≥ 14
	AlCl	CHN^2	AsH_3	C_2H_2O	C_2Cl_4	$C_2H_3NO^2$	$C_2H_4N_2$	$C_2H_5NO^2$	$C_2H_5NO_2$	C_2H_7NO	$C_3H_6O_3$	$C_3H_8O_2^2$	$C_{17}H_9N^3$
	AlF	CHP	C_2H_2	C_2H_2S	$C_2H_2O_2$	$C_2H_4O^3$	$C_2H_4O_2^4$	$C_2H_6O^2$	$C_2H_6O_2^2$	$C_3H_6O_2^2$	C_3H_7NO	$C_3H_9N^3$	$C_{16}H_{10}$
	Br_2	CO_2	$C_2N_2^2$	C_2HNO	C_2H_3Cl	$C_2H_4S^2$	$C_2H_5N^2$	$C_2H_6S^2$	C_2H_6OS	C_3H_7N	$C_3H_8O^3$	C_3H_9P	$C_{14}H_{10}$
	BrH	COS	CH_2O	C_3HN^2	$C_2H_3N^3$	$C_3H_2N_2$	C_2H_6	$C_3H_3N_3$	$C_2H_7N^2$	C_3H_8	$C_4H_6O_2^2$	$C_4H_8O^2$	$C_{11}H_{12}N_2O_2$
	C_2	CS_2	CH_2S	CCl_4	C_2H_4	C_3H_3N	$C_3H_3NO^3$	$C_3H_4N_2^2$	C_3H_5NO	C_4H_6O	$C_4H_7N^3$	C_4H_8S	$C_{11}H_7N^2$
	CO	ClNO	$CHNO^3$	CF_4	C_2HCl_3	$C_3H_4^3$	C_3H_3NS	C_3H_5N	$C_3H_6O^4$	C_5H_5N	$C_4H_8^5$	$C_5H_8^5$	$C_{10}H_8$
	CS	FNO	$CHNS^3$	CH_2Cl_2	$C_3H_2O^4$	C_3N_4	$C_3H_4O^2$	$C_3H_6^2$	$C_4H_4N_2^2$	C_5H_6	$C_6H_6^3$	C_7H_5N	C_9H_8
	Cl_2	H_2O	H_2O_2	CH_2F_2	C_4H_2	C_5HN	$C_4H_3N^3$	C_4H_4S	$C_4H_5N^2$		C_6N_6		$C_8H_{10}^3$
	ClH	H_2S	H_2S_2	$CH_2N_2^2$	C_4N_2	CH_4Si_2	C_4H_4	$C_5H_4^2$	$C_4H_6^4$				C_8H_8
	F_2	H_2Se	H_3N	CH_2O_2	CH_2O_3	CH_5N	C_6H_2						$C_7H_8^2$
	FH	HNO	H_3P	CH_2S_2	CH_3NO	N_2O_5	C_6N_2						C_6H_{10}
	H_2	N_2O	HNO_2	CH_3Br	CH_3NS		CH_3ClO_2S						$C_6H_{12}^9$
	N_2	O_2S	HNOS	CH_3Cl	CH_3NSi		CH_4N_2O						$C_6H_{14}^4$
	OSi	O_3	O_3S	CH_3F	CH_4O		H_6Si_2						C_6N_8
				CH_3I	CH_4S								$C_5H_{10}^3$
				CH_3P	H_4N_2								C_5H_{12}
				CH_4	N_2O_4								C_5H_8O
				$CHCl_3$									$C_4H_{10}^2$
				CHF_3									$C_4H_{10}O^4$
				GeH_4									$C_4H_{10}S$
				H_3NO									$C_4H_8O_2^3$
				H_4Si									$C_3H_8O_3$
													C_3N_{12}

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3.2 ECP impact on TICS calculation

As seen in Huo and Kim [19], enabling ECP for BEB calculations indeed increase the TICS for molecule containing $Z > 10$ elements. Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus

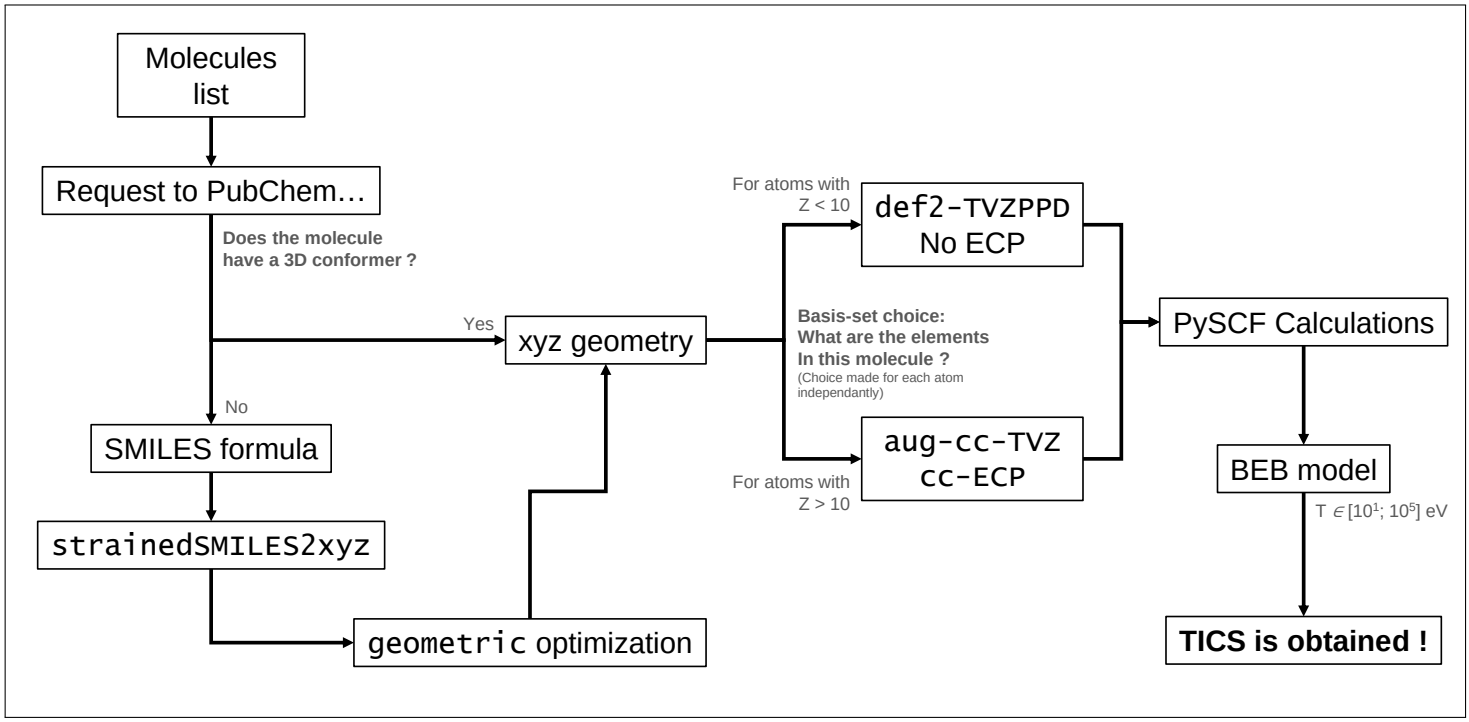


Figure 2: Overview of the general workflow and methodology employed in this work.

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Table 2: Mean Δ of TICS between ECP and non-ECP calculations for $3p$ and $4p$ elements bearing molecules.

	$\overline{\Delta\sigma}$ (%)	
Molecule	cc-ECP	CRENBL
H_3P	13.996	14.568
H_4Si	6.505	7.311
H_6Si_2	9.242	9.283
Geh_4	27.548	32.212
H_2Se	27.384	31.159
H_2S	18.474	16.958
Ch_3Br	20.059	22.418
Brh	24.582	27.666
Br_2	26.276	26.496
Cs	11.996	0.948
Cl_2O	18.123	12.592
Osi	5.703	6.007
Csi	7.079	7.372
Ch_4Si_2	5.093	7.995
Ch_3Nsi	2.749	3.753
$C_4H_{10}S$	9.545	9.718

4 Discussion and comparison with litterature

4.1 Overview

Des soucis avec les hydrocarbures saturés ? Le propene par ex

First and foremost, experimental B values tend to be smaller than theoritical/computed B values [2, 21]. As such, the cross section obtained through experimental B values tend to be higher,≈10-15% at the cross section peak. This was observed several time in the plots of figures 1-3, where theoritical curves were generally smaller than those from experimental papers. This is not systemic.

4.2 Limits

Whereas this paper serves as a database presentation, some results do harbour scientific interest, as some molecule haven't got a dedicated TICS paper yet. blalbalbalb
As shown in several studies, isomery play a minor role regarding σ_{TICS}. dshfdv

Ici évoque le SF6, qui a un petit intérêt astro mais que j'arrive pas a calcluer la géométrie

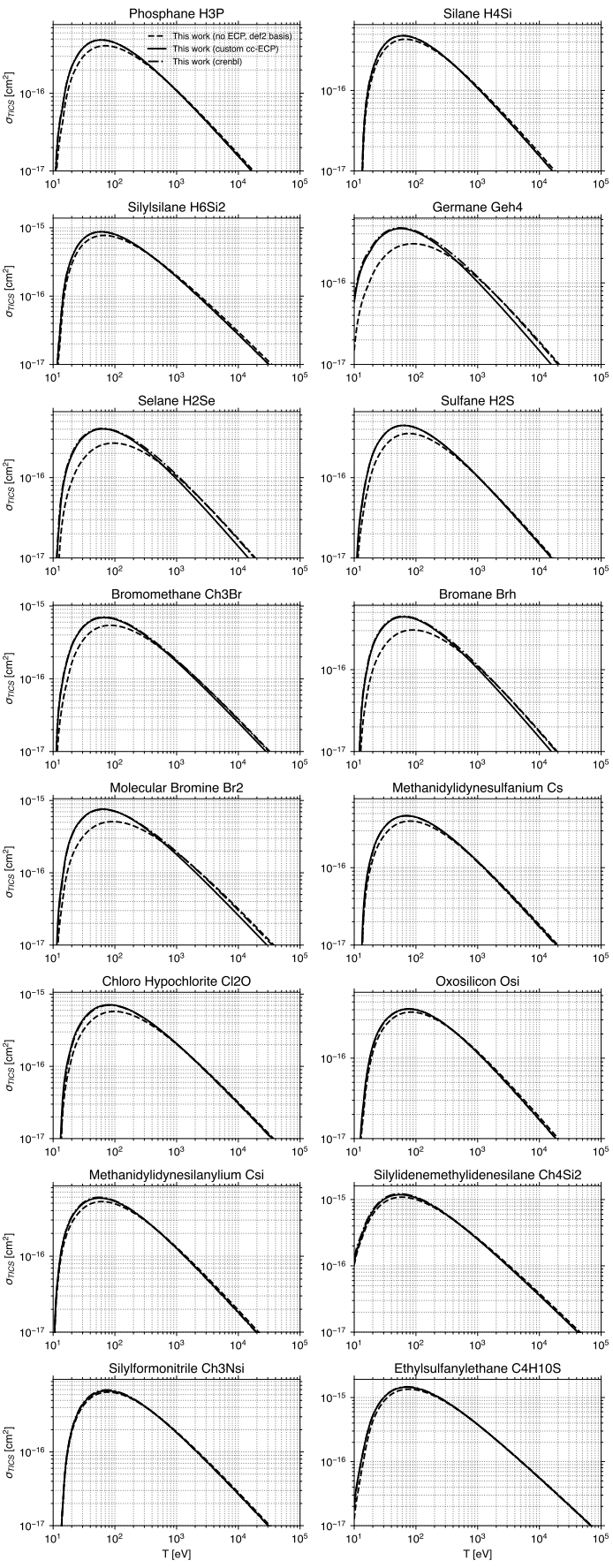


Figure 3: Comparison of TICS results for several molecule with and without ECP enabled

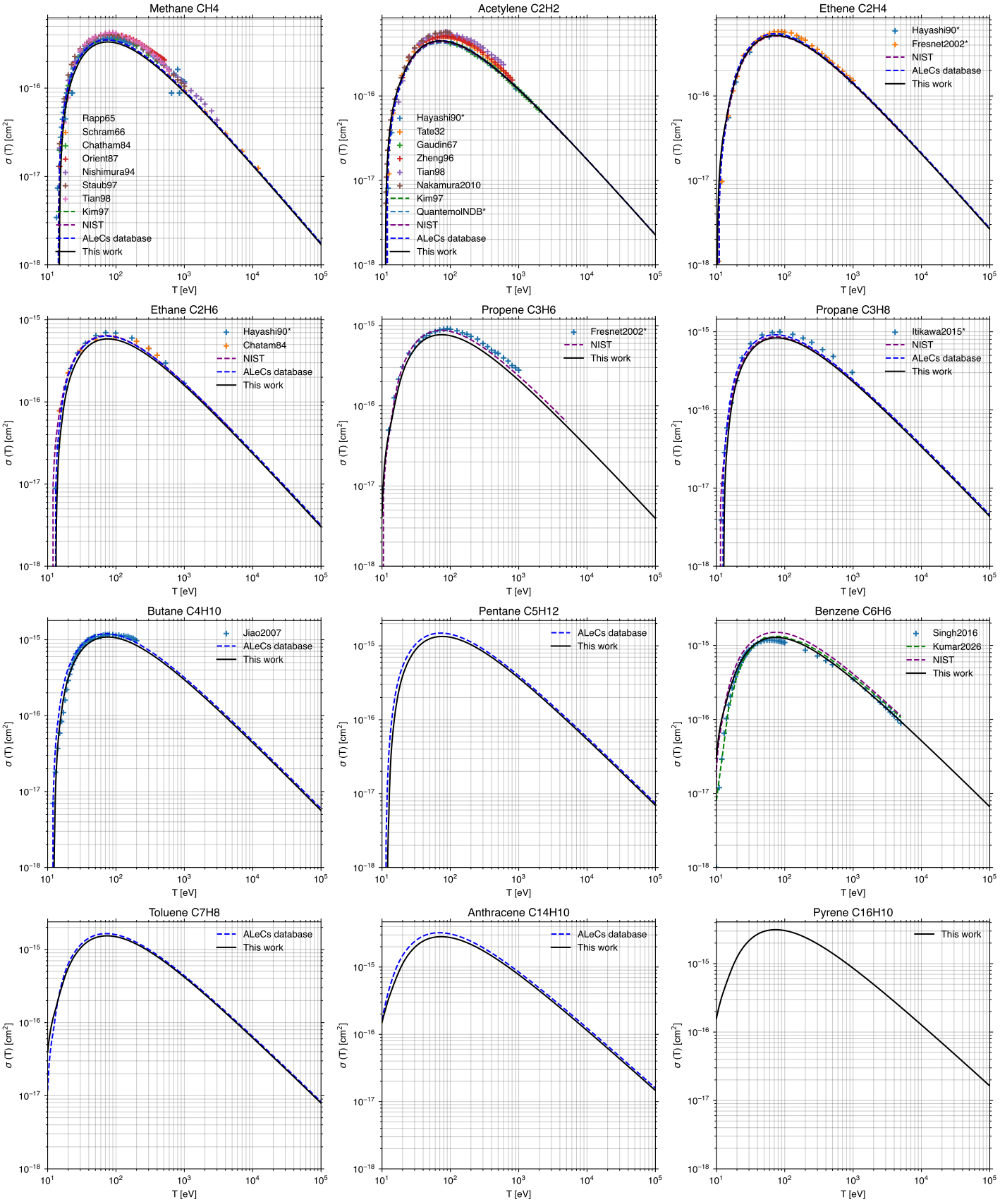


Figure 4: Calculted TICS for several hydrocarbons. In those plots, σ_{TICS} as been cropped at 10^{-8}. Scatter plot mean experimental data and line plot mean theoritical data. Those data come from Rapp and Englander-Golden [34], Jiao, DeJoseph, and Garscadden [22], Chatham et al. [8], and Ali et al. [2]. The references marked with an * where obtained through lxcat.com database [13, 7, 32, 31]. Those references include

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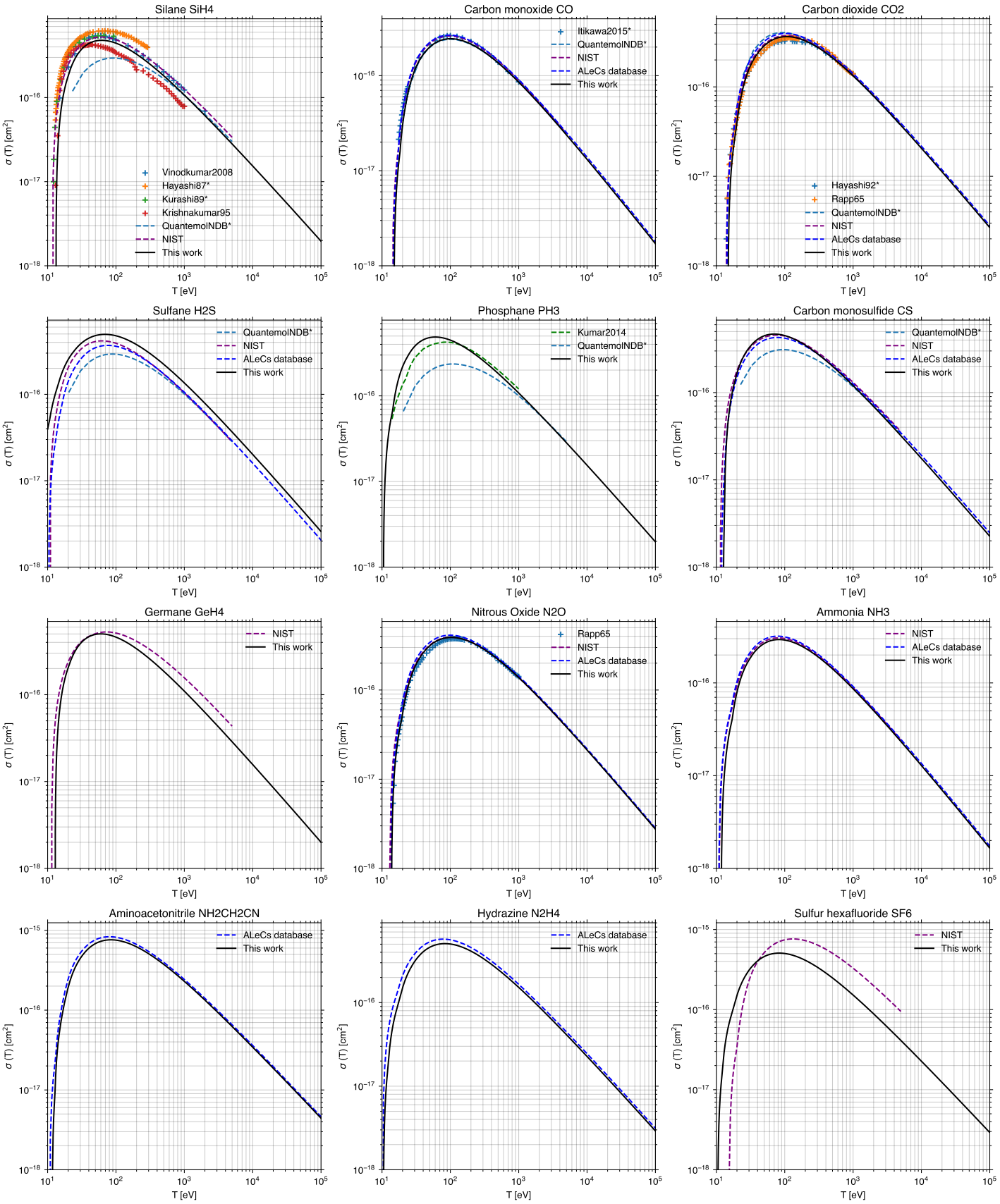


Figure 5: Calculted TICS for several organic and non-organic molecules. In those plots, σ_{TICS} as been cropped at 10^{-8}. Scatter plot mean experimental data and line plot mean theoritical data. Those data come from Rapp and Englander-Golden [34], Jiao, DeJoseph, and Garscadden [22], Chatham et al. [8], Ali et al. [2], and Kumar [26]. The references marked with an * where obtained through lxcat.com database. Those references include

5 Conclusion

Pouvoir utiliser un spectre complet de section efficace d'ionisation comme ceux que j'ai calculé pour calculer la section effiace d'ionisation partielle ? Pouvoir brancher cela a des autres modeles (comme celui de Gautier 2020) ? Faire du calcul de flux d'ionisation, jpeux essayer de lier ca a triton

Multiple outlook could be imaginated:

- Adapt the pipeline for radicals (*i.e* molecules with spin $\neq 0$) and ions; This would require to do calculation using the UHF method of SCF;
- Refine and optimize the geometry of molecules -if possible. This could be useful for molecules taken from a 2D structure.
- Adapt and create a custom `def2-` ECP basis for $3d$ and $4p$ elements in order to homogenize the database as much as possible protocol-wise;

Acknowledgement

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Appendix

Optimization of PubChem geometries